FlareLine quick start

version 0.1.0

Design aid only. All geometry must be verified and approved per your organization's engineering and quality process before fabrication.

Start

- Open a sample: File > Open sample > Simple 90. Save your own work with Ctrl+S (.flareline.json).
- Every start shows the disclaimer: FlareLine is a design aid only.

1 Tube spec 2 Ends

- Pick size, material and wall. Chips show where each value comes from: green VERIFIED, purple USER INPUT, grey DEFAULT, red UNVERIFIED / MISSING.
- Choose 37° flare, plain or flareless for each end. End points sit at the flare; the end allowance is added to the cut length.

3 Route & bends

- Edit the bend table: XYZ (points, Z up, P0 = end 1), LRA (Y feed between tangent points, B rotation, C bend angle) or Legs (point to point). All three stay in sync.
- LRA 'Marks from P0' gives start and end of each bend for hand bending.
- Add leg: Ctrl+L. Undo / redo: Ctrl+Z / Ctrl+Y. Rotation display (CW/CCW, viewpoint) is set in step 3.

4 Check

- Red errors: the tube cannot be made (overlapping bends, bend past the end, CLR <= OD/2, self-interference). Amber warnings: a limit or a data value needs attention.
- Click any number, 2D dimension or warning: the Explain tab shows its formula and full-precision inputs.

5 Drawing

- Fill in the title block, notes and revisions, then File > Export drawing (Ctrl+E): PDF, DXF, PNG, CSV on ANSI A, B or D.
- Unless there are no errors and all data is verified, the drawing is watermarked NOT FOR FABRICATION. The PDF includes a calculation record for every number.
- Approved deviations (Tools) replace a data value for one project, with approval reference and approver; they are printed in the notes.

Standards data

- Tools > Standards & shop data (Ctrl+D). Enter values from your licensed standards, give a reason, then Mark row verified with standard, revision, table and date. Every change is logged. FlareLine ships with no standards values.

Views and keys

- 3D: drag orbit, middle/Shift-drag pan, wheel zoom; F fit, 0 iso, 1 front, 2 top, 3 right, M measure, S section. 2D: wheel zoom, double-click fit. F1: How-to panel.